Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/05/2005
PI Name:	Beckingham, Kathleen M. Ph.D.		
Project Title:	Drosophila Behavior and Gene Expression in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	kate@rice.edu	**Fax:**	FY 713-348-5154
PI Organization Type:	UNIVERSITY	**Phone:**	713-348-4016
Organization Name:	Rice University Institute of Biosciences and Bioengineering		
PI Address 1:	6100 Main Street		
PI Address 2:	MS-140		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77005	**Congressional District:**	25
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/01/2002	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	NOTE: received NCE through 6/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharya, Sharmila (NASA Ames Research Center)		
Grant/Contract No.:	NCC2-1356		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Model organisms are of great value in deciphering the fundamental mechanisms underlying universal biological processes. The unparalleled genetic methodologies available for the fruit fly Drosophila melanogaster make it uniquely advantageous as a model system for the study of gravity perception. Drosophila has been adopted as the insect of choice for studies in the Insect Habitat of the International Space Station. The experiments proposed derive from the long-term objective of using Drosophila to understand the molecular processes of gravity perception. It is proposed to examine the effects of microgravity, and other gravitational force levels, on behavior and gene expression in Drosophila. Preliminary studies have identified a period of adaptation to hypergravitational force that is characterized by aberrant behavior, followed by resumption of normal activity. We will determine whether similar changes occur in microgravity and determine whether gene expression is altered during the critical adaptation period. The complete sequencing of the		

Task Description:	Drosophila genome and generation of microarrays representing essentially all genes within the organism will allow us to examine gene expression on a global level and identify all genes affected by gravitational force change. Preliminary work has also identified mutant Drosophila with abnormal gravitational sensing. Mutants for which the affected gene as been identified and proved i) to be novel ii) to be expressed in sense organs indicated to detect gravity and iii) to have a human homolog, will also be used in these studies. Determination of the gravity-induced behavioral and gene expression changes in these mutant files and comparison to the wild type parameters will address the fundamental roles of the affected genes in gravity perception and the relevance of any gene expression changes seen in wild type files.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A deeper understanding of the molecular mechanisms underlying graviperception.
Task Progress:	The two key components of the planned experiments on the International Space Station (ISS) are a) examine of the behavior of mutant and wild type strains of Drosophila during transitions between microgravity, 1g and 2 g force and b) collection of fly samples immediately before and after the gravity transitions to be used on return to Earth for microarray analysis of gene expression changes. During the past year, key preliminary experiments in preparation for the flight experiment have been performed. The flight experiment will study one wild type strain and three mutant strains of Drosophila that are defective in gravitaxic responses. Some of the year's experimentation was focused on studies to determine which of 24 gravitaxic mutants previously isolated would be optimal for use in the ISS experiment. Additional behavioral testing was performed to address the specificity of the defects in gravitaxic responses in all 24 strains. As a result of these studies, it was established that, in all but two of the mutants no other behavioral responses examined were altered. Molecular cloning studies were used to determine which genes are affected by each of these 24 mutations. It was thus established that both previously known genes and novel transcription units are affected by the mutations. This information will also be used to assess which mutants will yield the most valuable insights if selected for the ISS experiment. A final aspect of the gravitaxic mutant collection was also studied: the question of which peripheral sense organs on the adult integument are affected by the mutations. Interestingly, several sense organs of the head already implicated in mechanosensing appear to be affected in many of the mutations. Again, this information will be used in making the final mutant selection. Previously we have identified an unusual behavioral response in Drosophila upon subjection to hypergravity; after about 30 minutes, flies fall into a quiescent state for about 2 hours, which is then followed by a return to normal activity. We have investigated this response further since the signaling mechanisms underlying this response are likely to have direct relevance to the responses we will observe in microgravity. Studies of the hypergravity response in various mutant strains of Drosophila have established that a brain region termed the mushroom body is required for control of the response and that genes of the cAMP pathway act in the mushroom body to execute the response. A role for genes that regulate circadian rhythm in controlling hypergravity responses has also been uncovered. Further, mutants that increase the ability of flies to resist stress fail to show the hypergravity-induced quiescence. Gene expression studies using microarray technology have been used to address the molecular changes underlying the hypergravity responses. Gratifyingly, genes shown by the mutant studies described above to play a role in the response were found to undergo significant changes in expression during the response. Expression changes in many other signaling molecules were also detected by this work. It will be of great interest to compare the changes detected by these studies to those that occur in flies subjected to microgravity
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Abstracts for Journals and Proceedings	Beckingham KM. "A proper sense of gravity: Graviperception in insects." Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, November 2004. Gravitation and Space Biology Bulletin 2004 Nov;18(1):2. , Nov-2004
Abstracts for Journals and Proceedings	Texada MJ, Munjaal R, Armstrong JD, Beckingham KM. "Yuri, a gene affecting Drosophila gravitaxis." 45th Annual Drosophila Research Conference, Washington, DC, March 2004. 45th Annual Drosophila Conference Bulletin, p. 328, 726c, March 2004. , Mar-2004
Abstracts for Journals and Proceedings	Bhattacharya S, Fahlen TF, Sanchez ME, Stowers RS, Chin G, Shenasa M. "Responses to altered gravity in Drosophila." 45th Annual Drosophila Research Conference, Washington, DC, March 2004. 45th Annual Drosophila Conference Bulletin, p. 336, 749b, March 2004. , Mar-2004
Abstracts for Journals and Proceedings	Stowers RS, Sanchez ME, Fahlen TF, Shenasa M, Razaq A, Bhattacharya S. "The Drosophila locomotor response to hypergravity requires a functional mushroom body." Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, November 2004. Gravitational and Space Biology Bulletin 2004 Nov;18(1):56. , Nov-2004
Abstracts for Journals and Proceedings	Sanchez ME, Razaq A, Shenasa M, Stowers RS, Kakavand A, Bhattacharya S. "Development of the EMCS hardware for multigenerational growth of Drosophila melanogaster in space." Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, November 2004. Gravitational and Space Biology Bulletin 2004 Nov;18(1):63. , Nov-2004
Articles in Peer-reviewed Journals	Beckingham KM, Armstrong JD, Texada JM, Munjaal R, Baker DA. "Drosophila melanogaster: The model organism of of choice of the complex biology of multi-cellular organisms." Gravitational and Space Biology Bulletin, in press, November 2004. , Nov-2004
Articles in Peer-reviewed Journals	Armstrong JD, Texada MJ, Munjaal R, Baker DA, Beckingham KM. "Gravitaxis in Drosophila: a forward genetic screen." submitted to Journal of Neuroscience, November 2004. , Nov-2004
Books/Book Chapters	Beckingham KM, Texada MJ, Baker DA, Munjaal R, Armstrong JD. "Genetics of Graviperception in animals." in "Advances in Genetics." Elsevier Holland, in press, February 2005., Feb-2005